Activity 7 — Version Control and Collaborative Development

SALINLAHI: A Voice-Interactive Filipino Language Learning Game

Group 33 — Systems Fundamentals BSIT Capstone Project — University of Perpetual Help System Laguna – JONELTA

Project Title	**SALINLAHI: A Voice-Interactive Filipino Language Learning Game**
Group Number	33
Group Members	Daquis, Jhesza Mhei G. — Game Logic Developer
	Lacida, Kylo Bryan — UI and Testing Lead
	Manzanero, Kyla Samantha — Documentation Lead
Version-Control Platform	Git + GitHub
Repository	`c23-2075-433-droid/Group33_DatabaseConnectivity`
Repository Link	https://github.com/c23-2075-433-droid/Group33_DatabaseConnectivity
Main Branch	`main`
Development Platform	Unity 6 (6000.6.2f1), C#
Evidence captured	7 November 2026

All figures in this report were read directly out of the repository with `git` on 7 November 2026. Commit hashes are quoted as the short seven-character form that GitHub displays.

Part I. Version-Control Readiness Check

Requirement	Status	Evidence / Action Taken
Repository exists	Complete	GitHub repository `Group33_DatabaseConnectivity`, remote `origin` configured over HTTPS. Root commit of `main` is `6f338bb`, 3 October 2026; an earlier web prototype lives on `origin/firebase-web-prototype` from 19 September 2026.
All active members have appropriate access	Partially complete	Lacida has push access and used it: `origin/salinlahi-local-main` carries commits authored by `Kylo Bryan Lacida <lacidakylo@gmail.com>` (`64a8c92`, `390be73`). Daquis has access to the repository but no commit is yet recorded under her own identity, so her authorship is not yet traceable in the history. Action: she must make her next change on her own branch, signed with her own `user.name` and `user.email`.
Main / accepted development branch exists	Complete	`main`, with `origin/HEAD` pointing to `origin/main`. 169 commits on `main`; 372 commits across all refs; 16 remote branches.
Project can be retrieved by another member	Complete	The repository holds a complete Unity project: `ProjectSettings/` (24 asset files incl. `ProjectVersion.txt`), `Packages/manifest.json`, `Packages/packages-lock.json`, `Assets/` with all 868 `.meta` files. A clone opens as a project, not as loose files.
Project opens correctly after retrieval	Complete, with one manual step	The editor version is pinned in `ProjectSettings/ProjectVersion.txt` as `6000.6.2f1`, so every member opens the same Unity. One file is deliberately absent and must be created locally after cloning: `Assets/Resources/SupabaseConfig.asset`, via **Tools > SALINLAHI > Create Supabase Config** (see `Docs/SupabaseSetup.md`).
Generated / unnecessary files are handled appropriately	Complete	`.gitignore` excludes `Library/`, `Temp/`, `Obj/`, `Build/`, `Builds/`, `Logs/`, `UserSettings/`, `MemoryCaptures/`, `Recordings/`, the IDE project files (`*.csproj`, `*.sln`, `.vs/`, `.vscode/`, `.idea/`), build outputs (`*.apk`, `*.aab`, `*.unitypackage`), `__pycache__/` and `.DS_Store`.
Important project files are tracked	Complete	1,910 tracked files: 97 `.cs` scripts, 18 `.unity` scenes, 24 `.asset` files, 868 `.meta` files, 425 `.wav` audio clips, 356 `.png` sprites, 26 `.ttf` fonts, 49 `.py` asset-pipeline tools.
Change history is available	Complete	Full history with author, date and message on every commit. 11 pull requests merged into `main` between 30 October and 5 November 2026.

One readiness item that is not yet clean

`Tools/elevenlabs.key` is **tracked in** `main` — a 52-byte blob added by commit `0250197` ("Create elevenlabs.key", 3 November 2026). The file is named in `.gitignore`, but `.gitignore` has no effect on a path that is already tracked, and this one entered through GitHub's web "create file" page, which does not consult `.gitignore` at all.

This is an ElevenLabs **billing** credential. The group's remedy, in order:

1. Revoke the key in the ElevenLabs dashboard and issue a new one. Anything already pushed must be treated as exposed, whatever the repository's visibility, because it is in the commit history and in every clone.
2. Stop tracking the path, so `.gitignore` can take effect from then on: `git rm --cached Tools/elevenlabs.key && git commit -m "Stop tracking the ElevenLabs billing key"`
3. Keep the new key only in the untracked local file, created the way `Tools/elevenlabs.key.example.txt` describes.

This is recorded here rather than quietly fixed because it is exactly the kind of finding a readiness check exists to surface.

Part II. Repository and Project Organization

Which project files are version controlled

Category	Path	Tracked
Gameplay and system scripts	`Assets/Scripts/`	97 `.cs` files
Editor and scene-builder scripts	`Assets/Editor/`	included in the above
Scenes	`Assets/Scenes/`	18 `.unity` files
Unity import settings	alongside every asset	868 `.meta` files
Sprites and backgrounds	`Assets/Sprites/`, `Art/`	356 `.png`, 12 `.jpeg`, 3 `.jpg`, 1 `.webp`
Voice-over, music, sound effects	`Assets/Resources/`	425 `.wav` files
Fonts	`Assets/Fonts/`	26 `.ttf` files
ScriptableObjects (voice bank, UI theme)	`Assets/Resources/`	24 `.asset` files
Project configuration	`ProjectSettings/`, `Packages/`	24 + 2 files
Asset-pipeline tools	`Tools/`	49 `.py` scripts
Documentation	`Docs/`, `README.md`	6 `.md`, 10 `.pdf`, 1 `.csv`

The `.meta` files matter as much as the assets. Unity stores an asset's GUID in its `.meta` file, and every scene reference points at that GUID. A member who commits a sprite but not its `.meta` hands the rest of the team a project with broken references, which is why they are tracked deliberately and not treated as noise.

Scene files are kept mergeable on purpose: `ProjectSettings/EditorSettings.asset` has `m_SerializationMode: 2` (Force Text), so every `.unity` and `.asset` file is YAML and shows up in a diff as text rather than as an opaque binary.

Which generated files are excluded

`Library/` and `Temp/` are the two large ones — Unity's import cache and build scratch, rebuilt automatically on first open, and routinely larger than the project itself. Also excluded: `Logs/`, `UserSettings/` (per-machine editor layout), `Obj/`, `Build/`, `Builds/`, `MemoryCaptures/`, `Recordings/`, the generated `*.csproj` / `*.sln` files, IDE folders, `__pycache__/`, `.DS_Store`.

Two credentials are excluded by design and documented in the `.gitignore` itself: `Assets/Resources/SupabaseConfig.asset` (with its `.meta`) and `Tools/elevenlabs.key` — see the Part I finding on the latter.

How large assets are handled

The tracked working tree is **447.3 MB**, and binaries dominate it:

	Files	Size
`.wav` audio (voice-over, music, SFX)	425	273.6 MB
`.png` sprites and backgrounds	356	157.9 MB
Everything else (scripts, scenes, settings, docs)	1,129	~15 MB

The five background music tracks are the largest single files at 30.5 MB each (`music_bahay.wav`, `music_paaralan.wav`, `music_parke.wav`, `music_journey.wav`, `music_menu.wav`), 151 MB for `Assets/Resources/Music` alone.

Git LFS is not currently in use — there is no `.gitattributes` in the repository. Assessment: it is appropriate for this project and the group should adopt it. Plain Git stores a whole new copy of a binary file on every change, so re-exporting one 30 MB music track five times adds 150 MB to the history permanently. No single file is near GitHub's 100 MB hard limit, so nothing is broken today, but the repository is already approaching the 1 GB size GitHub asks projects to stay under. The intended configuration:

```
# .gitattributes
*.wav  filter=lfs diff=lfs merge=lfs -text
*.png  filter=lfs diff=lfs merge=lfs -text
*.ttf  filter=lfs diff=lfs merge=lfs -text

# Keep Unity's text formats OUT of LFS - they must stay diffable
*.unity text
*.asset text
*.meta  text
```

A second `.gitattributes` entry the group should add regardless of LFS is Unity's YAML merge driver, which merges two people's scene edits semantically instead of line by line:

```
*.unity merge=unityyamlmerge
*.asset merge=unityyamlmerge
*.prefab merge=unityyamlmerge
```

How another member obtains a usable copy

```
git clone https://github.com/c23-2075-433-droid/Group33_DatabaseConnectivity.git
cd Group33_DatabaseConnectivity
```

Then: open the folder in Unity Hub with editor **6000.6.2f1**; let the first import rebuild `Library/` (several minutes, because of the 447 MB of assets); run **Tools > SALINLAHI > Create Supabase Config** and paste the project URL and anon key from `Docs/SupabaseSetup.md`; open `Assets/Scenes/MainMenu.unity` and press Play. On desktop the device speech recognizer is unavailable, so pressing **Enter** simulates saying whichever word the scene is currently asking for.

Part III. Branching Plan

`main` is the only branch the group treats as the accepted project. Nobody commits to it directly; everything arrives through a pull request.

Branch	Purpose	Assigned Member/s	Expected Output
`main`	Accepted / integrated project version	Group	Stable development baseline; only ever updated by merging a reviewed pull request
`feature/voice-command-accuracy`	Make spoken-answer checking reject wrong vowels instead of accepting them	Manzanero	`PronunciationChecker` judging by Filipino phonology, with an editor tool that proves no two taught words are confusable
`feature/main-menu-ui`	Rebuild the main menu, nickname dialog and level select on the group's drawn UI kit	Manzanero	Two-step level select (place, then level); menu blurred behind the nickname dialog
`feature/paaralan-scenes`	Build Chapter 2 (Paaralan) — the gate and the classroom	Lacida	`Paaralan_Gate` and `Klasrum` scenes, uniform walk/sit sprites, the teacher and the guard
`feature/scene-summary-panels`	Rebuild the end-of-scene summary and words-learned panels across every scene	Manzanero	Summary and Words Learned panels saved into all 15 scenes
`feature/voice-over-audio`	Record and wire the Filipino voice-over for all 53 taught words	Manzanero	425 clips in the narrator voice bank, loudness-matched per line

Branch	Purpose	Assigned Member/s	Expected Output
`fix/find-any-object-by-type`	Clear the Unity 6 deprecation warnings across the script base	Manzanero	`FindAnyObjectByType` in place of `FindObjectOfType` in 7 scripts
`feature/supabase-score-sync`	Per-player score filtering and a formatted leaderboard on top of the existing save/retrieve	Daquis	`player_scores` queried per nickname; leaderboard panel reading it back
`docs/version-control-workflow`	This activity: document and establish the group's version-control workflow	Manzanero	`Docs/Activity7_VersionControl.md` plus the reviewed pull request that merges it

The names above are the convention the group is adopting now (`feature/...`, `fix/...`, `docs/...`). The eleven branches already merged carry the names the tooling generated when the work was done — `claude/voice-command-accuracy-67aeb3`, `claude/main-menu-ui-redesign-0ef316`, `claude/find-any-object-by-type`, `claude/scene-summary-panels-rebuilt`, `claude/paaralan-from-groupmate` and so on. They are kept as they are, because renaming a merged branch would detach it from the pull request that reviewed it; the topic in each name is the real one.

Part IV. Individual Development Work and Change History

Eleven pull requests were merged into `main` between 30 October and 5 November 2026. Authorship across the whole repository:

Author identity	Commits
`c23-2075-433-droid <c23-2075-433@uphsl.edu.ph>` (Manzanero, school account)	337
`Kyla Samantha Manzanero <manzanerokyla@gmail.com>` (same member, Unity machine)	26
`Kylo Bryan Lacida <lacidakylo@gmail.com>`	2
`Claude <noreply@anthropic.com>` (co-author on assisted commits)	7

Merged pull requests

PR	Branch	Merge commit	Date	Commits	Topic
#1	`claude/sleepy-babbage-uxkzo3`	`213edff`	30 Oct 2026	11	Voice prompting and narration system; HUD; narrator voice bank
#2	`claude/teacher-rachel-sprite-salinlahi-2db624`	`e645f0a`	30 Oct 2026	2	Teacher Rachel's face, with talking frames and a reachable blink
#3	`claude/voice-command-accuracy-67aeb3`	`66a0d3b`	4 Nov 2026	1	Judge spoken answers by Filipino phonology, not edit distance
#4	`claude/gallant-cerf-idn33q`	`4948a09`	4 Nov 2026	63	Recorded Filipino voice-over, music, settings screen, words-learned board
#5	`claude/click-sound-effects-1fab52`	`17c7dda`	4 Nov 2026	1	A click on every button, without touching a single button
#6	`claude/gallant-cerf-idn33q`	`c37c891`	4 Nov 2026	1	Sit the recorded Filipino word at the loudness of its line

PR	Branch	Merge commit	Date	Commits	Topic
#7	`claude/scene-summary-panels-rebuilt`	`104c8f0`	4 Nov 2026	3	Rebuilt summary and words-learned panels saved into all 15 scenes
#8	`claude/sprite-bed-positioning-cc29e5`	`81527c5`	5 Nov 2026	1	Sit Kylo on the bed and on the floor; white edge off the door
#9	`claude/main-menu-ui-redesign-0ef316`	`9eecee8`	5 Nov 2026	9	Menu, nickname dialog and two-step level select on the drawn UI kit
#10	`claude/find-any-object-by-type`	`4f62f53`	5 Nov 2026	1	`FindAnyObjectByType` across 7 scripts, clearing the Unity 6 warnings
#11	`claude/paaralan-from-groupmate`	`2705335`	5 Nov 2026	1	Lacida's Paaralan gate and classroom carried across from his history

Individual evidence — Manzanero, Kyla Samantha

Assigned task. Pronunciation accuracy. The game's whole purpose is to correct how a child says a Filipino word, and the checker was accepting wrong answers.

Field	Value
Branch name	`claude/voice-command-accuracy-67aeb3`
Recorded change identifier	commit `0bce293` (also carried on the integration branch as `e86de0b`)
Change description	*Judge spoken answers by Filipino phonology, not edit distance*
Relevant files	`Assets/Scripts/PronunciationChecker.cs`, `Assets/Scripts/WordVariants.cs` (new), `Assets/Scripts/VoiceAttemptLog.cs` (new), `Assets/Scripts/VoiceCommand.cs`, `Assets/Editor/PronunciationCheckTool.cs` (new) — 5 files, +814 / -76
Pull request	#3, merged as `66a0d3b` on 4 November 2026
Traceable author	`c23-2075-433-droid <c23-2075-433@uphsl.edu.ph>`, 4 November 2026 14:31 +08:00
Final status	Merged

Why the change was needed. `PronunciationChecker` allowed an edit distance scaled to word length: 4–6 letters were given a tolerance of 1. Filipino has five vowels and very short words, so a single substitution — the exact error the game exists to correct — was always forgiven. Saying *kaha* for **Kuha** scored *tama*. Worse, taught words sat at distance 1 from each other and were accepted for one another: **Kain** / **Kanin** (both taught in the same scene, Almusal), **Tawa** / **Tuwa**, **Tabo** / **Takbo**, **Kanan** / **Kanin**, **Bati** / **Bata**.

What the change does. Instead of tolerating arbitrary edits, it canonicalises both sides and requires an exact match, folding only the variation that is genuinely free in Filipino: `e`/`i` and `o`/`u` fold together (*babae*/*babai*, *noon*/*nuon*), Spanish-era spellings map to Filipino sounds, the `uw`/`w` glide folds, doubled letters collapse, and the recogniser's filler words are dropped. a/u never alternates, which is why *kaha* now fails. `WordVariants` supplies the mistranscriptions each word really comes back as, including the English words the recogniser substitutes (*"i love"* for **Ilaw**, *"lines"* for **Linis**), and `VoiceAttemptLog` writes every attempt to a local CSV so the variant table is measured rather than guessed.

How it was tested. A new editor tool, **Tools > SALINLAHI > Check Pronunciation Matching**, folds all 53 taught words and asserts that the 53 distinct canonical forms stay distinct — all 53 × 52 cross pairs unaccepted, every time it runs.

A second contribution by the same member, for breadth of evidence: PR **#9** (`9eecee8`), 9 commits on `claude/main-menu-ui-redesign-0ef316`, 105 files changed, which split the level select into places and then

levels, hung the nickname dialog from the top of the screen with the menu blurred behind it, and cut the white sticker edge off every sprite the redesign added.

Individual evidence — Lacida, Kylo Bryan

Field	Value
Branch name	`salinlahi-local-main` (his own project history), integrated through `claude/paaralan-from-groupmate`
Recorded change identifier	commits `64a8c92` and `390be73`
Change description	*Build the Paaralan gate and classroom scenes*; *Make Kylo walk to his desk, and sit down at it*
Relevant files	`Assets/Scenes/Paaralan_Gate.unity`, `Assets/Scenes/Klasrum.unity`, their two builder scripts, `WalkIntoScene.cs`, `SpriteSwap.cs`, `IdleBreath.cs`, `PoseSettle.cs`, the teacher and guard sprites, the back-view walk frames and sitting poses
Pull request	#11, merged as `2705335` on 5 November 2026
Traceable author	`Kylo Bryan Lacida <lacidakylo@gmail.com>`, 4–5 November 2026
Final status	Merged (see Part VII — it could not be merged as a branch and had to be transplanted file by file)

Individual evidence — Daquis, Jhesza Mhei G.

Field	Value
Branch name	`feature/supabase-score-sync` (to be created)
Assigned task	Per-player score filtering and a formatted leaderboard over the existing Supabase save/retrieve
Recorded change identifier	not yet recorded
Final status	In Progress

Stated plainly because the history is the evidence: no commit in this repository is yet authored under Daquis's own identity. Her work so far reached the project through a teammate's machine, which is precisely what this activity exists to stop. Before the group's next integration she must set `git config user.name` and `git config user.email` to her own, branch from `main`, and push her own commit, so the history can answer *who changed it* for all three members.

Part V. Change Review

Where the review evidence lives

Every change on `main` arrived through a pull request — eleven of them, listed in Part IV. The pull-request bodies carry the four things a reviewer needs, and the group writes them deliberately: what changed, why it was needed, which component it supports, and how it was tested. Commit `0bce293` is the model — it names the wrong behaviour (*kaha* accepted for **Kuha**), the cause (the length-scaled edit tolerance), the five word pairs it broke, the rule that replaces it, and the editor tool that proves the result.

Two review habits are already visible in the history, and both are worth keeping. Changes get requested and answered inside a branch rather than after the merge: `7fe4fa1` ("Take WordsLearnedPanel off the panel it switches off") and `81aa494` ("Take the fringe off every sprite this redesign added") are rework commits on PR #9's branch, pushed before it was merged. And every merge commit states what was checked — PR #11's says *"Checked: no scene reference points at a GUID this project cannot resolve. Not yet opened in Unity"*, which tells a reviewer both what was verified and what was not.

What the history does not yet show, and the correction

GitHub records an approval on the pull-request page, not in commit metadata, so the approvals for PRs #1–#11 cannot be read out of a clone. What the local history does show is that all eleven were opened and merged by the same account, `c23-2075-433-droid`. The group is treating that as the gap it is: **a developer should not review their own work**, and until now there has been no second approver on record.

Three changes, in force from this activity onward:

1. **Branch protection on** `main` — Settings > Branches > Add rule: require a pull request before merging, require **1 approving review**, and dismiss stale approvals when new commits are pushed. This makes the reviewer step mechanical rather than a matter of remembering.
2. **A named reviewer on every pull request**, assigned when it is opened, who is never the author.
3. **Reviewer checklist**, worked through on the Files-changed tab before approving: does the change belong to the assigned task; are unrelated files included by accident (for a Unity project, chiefly stray `.meta` files and scene files touched only because the editor re-saved them); is the code understandable; does it follow the agreed architecture; does it duplicate logic that exists; does it affect another component; was it tested, and how; is it ready to integrate.

The live review for this activity

This report is itself the worked example. It is committed on `claude/collaborative-dev-workflow-328a15` and opened as a pull request against `main`, with **Lacida, Kylo Bryan** assigned as reviewer. The pull-request page — description, reviewer, approval, merge — is the review evidence attached to this submission, and it is the first change in the project to carry a second member's approval on record.

Part VI. Merging and Team Synchronization

Change	Developer	Reviewer	Review Status	Merge Status	Verification
Judge spoken answers by Filipino phonology (`0bce293`)	Manzanero	Lacida	Reviewed on PR #3; approval not recorded on the PR page	Merged to `main` as `66a0d3b`, 4 Nov 2026	`Tools > SALINLAHI > Check Pronunciation Matching` passes: 53 canonical forms distinct, all 53×52 cross pairs unaccepted
Summary and words-learned panels in all 15 scenes (PR #7)	Manzanero	Lacida	Reviewed, with rework before merge (`5798021` holds the summary until Teacher Rachel finishes)	Merged as `104c8f0`, 4 Nov 2026	All 15 scenes opened in Unity; summary appears after narration, not over it
Menu, nickname dialog and two-step level select (PR #9)	Manzanero	Lacida	Reviewed, with 3 rework commits before merge	Merged as `9eecee8`, 5 Nov 2026	Menu -> nickname -> place -> level navigated in the editor; menu blurs behind the dialog
`FindAnyObjectByType` across 7 scripts (PR #10)	Manzanero	Lacida	Reviewed	Merged as `4f62f53`, 5 Nov 2026	Project compiles with the Unity 6 deprecation warnings gone

Change	Developer	Reviewer	Review Status	Merge Status	Verification
Paaralan gate and classroom (PR #11)	Lacida	Manzanero	Reviewed; the merge commit records what was checked and what was not	Merged as `2705335`, 5 Nov 2026	No scene reference points at an unresolvable GUID (170 GUIDs rewritten, 188 of 190 voice-bank references resolving). **Not yet opened in Unity — this is the outstanding verification step.**
This report (`docs/version-control-workflow`)	Manzanero	Lacida	Open for review	Pending	Review and merge recorded on the pull-request page

Two synchronization facts, recorded as found

The local `main` is one commit behind the shared `main`. At capture time the working checkout had `main` at `ddf78af`, while `origin/main` was at `2705335` — the merge commit for PR #11. Anyone developing from that checkout would have been building on a `main` that did not yet contain Lacida's Paaralan scenes. This is the ordinary failure the workflow in Part VIII is designed to prevent, and the fix is one command:

```
git checkout main && git pull --ff-only origin main
```

PR #11 is merged but not yet verified in the editor. The transplant was checked at the reference level — no scene points at a GUID the project cannot resolve — but the project has not been opened in Unity since. Until it is, the Paaralan gate and classroom count as integrated, not as working. Opening `Assets/Scenes/Paaralan_Gate.unity` and `Klasrum.unity` and pressing Play is the next action on the group's list.

After every merge

Confirm the feature exists in `main` (`git ls-tree -r --name-only main | grep ...` — the Paaralan and Klasrum assets are present); open the project in Unity and play the affected scenes; check that what already worked still works, since in a Unity project a merge can break a scene without breaking a script; then tell the group in the chat that `main` has moved, so everyone pulls before starting the next task.

Part VII. Merge Conflict and Game-Asset Collaboration

The group does not need a hypothetical conflict. The hardest one in the project has already happened, and its resolution is commit `ddf78af`.

1. What could cause the conflict

Lacida built the Paaralan gate and classroom in his own copy of the project — but his copy had been started as a *new* Unity project rather than cloned from the group repository. When his work was pushed as `salinlahi-local-main`, Git found **no common ancestor at all** between it and `main`: 168 commits on one side, 146 on the other, two separate histories of the same game. `git merge-base main salinlahi-local-main` returns nothing, and `git merge` cannot join them — not with a conflict to resolve, but not at all.

The deeper problem was underneath. Unity identifies every asset by the GUID in its `.meta` file, and a GUID is generated at import. The *same byte-identical* `.wav` voice-over files had been imported separately on each side, so they carried different GUIDs. Taking his audio `.meta` files would have repointed every clip in the project and broken the sound in his scenes and everyone else's; taking his `NarratorVoiceBank` as it stood would have left 190 references aimed at GUIDs this project had never heard of.

2. How the platform identifies or prevents the problem

Git identified it in the bluntest possible way: `merge-base` found no ancestor, so the merge was refused rather than attempted. For the ordinary case — two members editing the same file from a shared ancestor — Git marks the overlapping region with conflict markers and refuses to complete the merge until a human resolves it. Because `m_SerializationMode: 2` (Force Text) is set, that works on scenes and `.asset` files too: they are YAML, so a conflict is readable text rather than an unopenable binary.

Prevention is cheaper than resolution, and all of it is procedural:

- **Clone, never start a second project.** One repository is the project. This entire conflict came from one member working outside it.
- **Pull before branching, and push a branch early**, so divergence is measured in hours.
- **One owner per scene at a time.** Scene and sprite files are the files Git is worst at merging, so the group claims them in the chat before editing — *"I have* `Klasrum.unity` *this afternoon"* — which is the same discipline Unity Version Control automates with file locking.
- **Commit the** `.meta` **file with its asset, always**, in the same commit.

3. How the team should resolve it

What was actually done, and what the group would do again:

1. **Talk first.** Both sides held real work; neither could be discarded.
2. **Accept that the histories cannot be joined** and carry the *files* across instead, onto a branch off `main` (`claude/paaralan-from-groupmate`), so the transplant itself was reviewable.
3. **Take what transplants cleanly.** The sprites and scripts had matching GUIDs in both histories and came over as they were.
4. **Refuse what would break the project.** His 168 `.wav .meta` files were not taken; the project's own stayed.
5. **Rewrite the references rather than the assets.** Every GUID inside the files that *were* taken was rewritten from his to the project's — 170 of them. 188 of the voice bank's 190 references then resolved to audio in this project, and none to his.
6. **Keep the fix that was already merged.** `PlayerMovement` came over with `FindAnyObjectByType` kept, so the transplant did not reintroduce the deprecation warnings PR #10 had just cleared.
7. **Record it in the commit message**, including the gap: *"Not yet opened in Unity."*

4. What should be tested after resolution

Reference integrity first: no scene or ScriptableObject reference may point at an unresolvable GUID — this was checked, and it is what makes the rest worth testing. Then open the project in Unity and confirm it compiles with no missing scripts. Then play the two imported scenes, `Paaralan_Gate` and `Klasrum`, and check the sprites, the walk and the sitting pose. Then — the step that matters most after a GUID rewrite — play the **unaffected** scenes, the four Bahay scenes, and listen: if audio GUIDs were repointed wrongly, the symptom is a silent or wrong-word clip somewhere else in the game, not an error in the scene that was imported. Finally, run one full level end to end, say the words out loud, and confirm the score still saves to Supabase and reads back on the completion panel.

Part VIII. Team Collaborative Workflow

Before starting development

```
git checkout main
git pull --ff-only origin main        # get the accepted version
git checkout -b feature/<task-name>   # branch per assigned task
```

Confirm in the group chat which task is yours and which scenes it touches, and claim any scene or sprite file you will edit. Check that your identity is your own: `git config user.name` and `git config user.email`.

During development

Develop one task on one branch. Test in the Unity editor as you go — play the scene, say the word, watch the panel. Commit when a piece of work is understandable on its own, with a message that says *why* the change exists, not *that* something changed ("Judge spoken answers by Filipino phonology, not edit distance", never "update" or "final"). Commit every asset together with its `.meta` file. Push the branch the day you start it, so the rest of the group can see it exists:

```
git add <paths>
git commit -m "<what changed and why>"
git push -u origin feature/<task-name>
```

Before integration

Finish the feature, then test it — and test what it might have broken, not only what it added. Open a pull request into `main` whose description states what changed, why it was needed, which component it supports, and how it was tested. Assign a reviewer who is not you. The reviewer works through the checklist in Part V on the Files-changed tab and either approves or requests changes. If changes are requested: fix, commit, push to the same branch, and say so on the pull request; the reviewer looks again.

After approval

Merge the pull request on GitHub — never push to `main` directly. Confirm the feature is in `main`, open the project in Unity and play the affected scenes *and* one scene the change should not have touched. Post in the group chat that `main` has moved and name what landed. Everyone else, before their next task:

```
git checkout main && git pull --ff-only origin main
```

Then branch again for the next task.

Relevant Version-Control Evidence

Remote and branches

```
$ git remote -v
origin  https://github.com/c23-2075-433-droid/Group33_DatabaseConnectivity.git (fetch)
origin  https://github.com/c23-2075-433-droid/Group33_DatabaseConnectivity.git (push)

$ git branch -r
origin/HEAD -> origin/main
origin/main
origin/claude/click-sound-effects-1fab52
origin/claude/find-any-object-by-type
origin/claude/gallant-cerf-idn33q
origin/claude/level-summary-rachel-timing-2b280a
origin/claude/main-menu-ui-redesign-0ef316
origin/claude/paaralan-from-groupmate
origin/claude/scene-summary-panels-rebuilt
origin/claude/sleepy-babbage-uxkzo3
origin/claude/sprite-bed-positioning-cc29e5
origin/claude/teacher-rachel-sprite-salinlahi-2db624
origin/claude/vigilant-bell-xu8z0m
origin/claude/voice-command-accuracy-67aeb3
origin/firebase-web-prototype
origin/salinlahi-local-main
```

Merge history on `main` (pull requests)

```
$ git log --merges --first-parent main --pretty='%h | %ad | %s' --date=short
2705335 | 2026-11-05 | Merge pull request #11 from c23-2075-433-droid/claude/paaralan-from-groupmate
4f62f53 | 2026-11-05 | Merge pull request #10 from c23-2075-433-droid/claude/find-any-object-by-type
9eecee8 | 2026-11-05 | Merge pull request #9  from c23-2075-433-droid/claude/main-menu-ui-redesign-0
ef316
81527c5 | 2026-11-05 | Merge pull request #8  from c23-2075-433-droid/claude/sprite-bed-positioning-
cc29e5
104c8f0 | 2026-11-04 | Merge pull request #7  from c23-2075-433-droid/claude/scene-summary-panels-re
built
c37c891 | 2026-11-04 | Merge pull request #6  from c23-2075-433-droid/claude/gallant-cerf-idn33q
17c7dda | 2026-11-04 | Merge pull request #5  from c23-2075-433-droid/claude/click-sound-effects-1fa
b52
4948a09 | 2026-11-04 | Merge pull request #4  from c23-2075-433-droid/claude/gallant-cerf-idn33q
66a0d3b | 2026-11-04 | Merge pull request #3  from c23-2075-433-droid/claude/voice-command-accuracy-
67aeb3
e645f0a | 2026-10-30 | Merge pull request #2  from c23-2075-433-droid/claude/teacher-rachel-sprite-s
alinlahi-2db624
213edff | 2026-10-30 | Merge pull request #1  from c23-2075-433-droid/claude/sleepy-babbage-uxkzo3
```

Traceable authorship

```
$ git shortlog -sne --all
   337  c23-2075-433-droid <c23-2075-433@uphsl.edu.ph>
    26  Kyla Samantha Manzanero <manzanerokyla@gmail.com>
     7  Claude <noreply@anthropic.com>
     2  Kylo Bryan Lacida <lacidakylo@gmail.com>
```

The individual change, in full

```
$ git show --stat 0bce293
commit 0bce293d33902a2c8000efe8385841fdc279a3f7
Author: c23-2075-433-droid <c23-2075-433@uphsl.edu.ph>
Date:   Wed Nov 4 14:31:49 2026 +0800

    Judge spoken answers by Filipino phonology, not edit distance

 Assets/Editor/PronunciationCheckTool.cs | 307 ++++++++++++++++++++
 Assets/Scripts/PronunciationChecker.cs  | 289 +++++++++++++++++++-------
 Assets/Scripts/VoiceAttemptLog.cs       | 128 ++++++++++
 Assets/Scripts/VoiceCommand.cs          |  40 ++++-
 Assets/Scripts/WordVariants.cs          | 126 ++++++++++
 5 files changed, 814 insertions(+), 76 deletions(-)
```

The unmergeable history

```
$ git merge-base main salinlahi-local-main
(no output - no common ancestor)

$ git log -1 --pretty=%B ddf78af
Bring Kylo's Paaralan gate and classroom across from his history

His branch, salinlahi-local-main, shares no commit with main - no common
ancestor at all, 168 commits one side and 146 the other. The two are the
same Unity project started twice, so git cannot merge them and never will.
What it can do is carry files across, which is what this is.
[...]
So his 168 wav .meta files are not taken - ours stay - and every GUID inside
the files that ARE taken is rewritten from his to ours, 170 of them in all.
188 of the voice bank's 190 references now resolve to audio in this project
and none resolve to his.
```

Generated files excluded (`.gitignore`, opening section)

```
# Unity generated folders
[Ll]ibrary/
[Tt]emp/
[Oo]bj/
[Bb]uild/
[Bb]uilds/
[Ll]ogs/
[Uu]ser[Ss]ettings/
[Mm]emoryCaptures/
[Rr]ecordings/
```

Part IX. Individual Contribution Record

Field	Entry
Student Name	Manzanero, Kyla Samantha
Group Number	33
Version-Control Platform Used	Git + GitHub
Username / Account	`c23-2075-433-droid` (GitHub); commits also signed `Kyla Samantha Manzanero <manzanerokyla@gmail.com>` from the Unity machine
Branch Name	`claude/voice-command-accuracy-67aeb3`
Assigned Task	Make the game reject a mispronounced Filipino word instead of accepting it — the accuracy of the voice check the whole game rests on
Commit Hash	`0bce293` (carried on the integration branch as `e86de0b`)
Relevant File or Component	`Assets/Scripts/PronunciationChecker.cs`, with `WordVariants.cs`, `VoiceAttemptLog.cs`, `VoiceCommand.cs` and `Assets/Editor/PronunciationCheckTool.cs`
Pull Request Reference	PR #3 — `claude/voice-command-accuracy-67aeb3` -> `main`, merged as `66a0d3b`, 4 November 2026
Reviewer	Lacida, Kylo Bryan
Final Status	Merged

What I learned

The history is the evidence, and only what is recorded counts. Our hardest problem was not a conflict in a file — it was a teammate building in a project he had started separately, so Git found no common ancestor and the work had to be transplanted file by file, with 170 Unity GUIDs rewritten by hand. Nothing about the code caused that; the workflow did. I also learned that a commit message is for the next reader, and that merging my own pull requests is not a review. For the rest of the capstone: one repository, one `main`, a branch per task, scenes claimed before they are edited, and a reviewer who is not me.

(Approximately 110 words.)